End-to-End Deep Portfolio Optimization with Risk- and Turnover-Aware Objectives

ARIN5203 Project Milestone Report

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1. Introduction

A portfolio model can either forecast returns and pass the forecasts to an optimizer, or output allocation weights directly and be trained on portfolio performance. The second, end-to-end approach avoids feeding forecast errors into the optimizer [1, 2], but its training objective then decides what kind of portfolio the network learns. Our project studies that choice. We train a small Transformer allocator with a return objective and add a variance penalty, an L1 turnover penalty, or both. We ask whether the penalties lower volatility and trading, whether the benefit survives transaction costs, and how much return they give up versus unregularized training and simple rules such as equal weighting [3]. The contribution is a controlled study of objectives, not a new architecture.

Plan and progress. The plan has four stages: (1) build a leakage-free dataset and a cost-aware backtester; (2) implement the baselines and the four objective variants; (3) select penalty strengths and training length on a validation period; (4) evaluate the frozen models once on a held-out test period with paired statistical tests. All four stages have been run, including the frozen-model test backtest (2021–2025) and the primary bootstrap; this report focuses on validation results (2018–2020) and leaves test analysis to the final report. Test data were not used for fitting, scaling or selection.

2. Problem Statement

Dataset. Daily adjusted open and close prices for six ETFs, SPY, EFA, EEM, IEF, TLT and GLD (U.S., developed and emerging equities, intermediate and long Treasuries, gold), downloaded from Yahoo Finance with yfinance [6]. The study covers 2006–2025 (5,031 common trading days); 2005 is used only to warm up features. The series share one calendar and have no missing, duplicated or non-positive prices. We did not fill or winsorize any values. The risk-free rate is the FRED 3-month T-bill rate (DTB3) [9], converted from its discount quote to a daily return and applied with a lag. All results use one team snapshot (downloaded 3 Nov. 2026, yfinance 1.2.0), frozen by SHA-256 hashes.

Splits. Training 2006–2017 (626 weekly decisions), validation 2018–2020 (157), test 2021–2025 (261). Feature scaling is fitted on training data only. Holding-period targets are cut at each split boundary, so no training target uses validation prices and no validation target uses test prices.

Task. At each weekly decision the model receives 60 trading days of 18 features (daily return, 20-day volatility and 20-day momentum per asset) and outputs six long-only weights summing to one. The signal uses data up to the previous close; trades execute at the next open, and holdings drift until the following rebalance.

Evaluation. The primary metric is the annualized net Sharpe ratio of daily returns in excess of the risk-free rate at 10 bps per traded dollar. We also report net CAGR, volatility, maximum drawdown (MDD) and annualized L1 turnover, each at 0, 10 and 25 bps. Test comparisons use a paired studentized circular-block bootstrap [4].

Expected results. The variance penalty should lower volatility and drawdown, the turnover penalty should lower trading and cost drag, and together they should give the best net Sharpe, at the price of some raw return.

3. Technical Approach

Allocator. Each day's 18 features are linearly projected to $d = 32$ and given sinusoidal position encodings. A 2-layer Transformer encoder (4 heads, dropout 0.1) processes the window, and the last token passes through LayerNorm, a linear layer and a softmax to give the weights (17,958 parameters, trained from scratch). The window contains only past data, so bidirectional attention inside it does not leak future information. An LSTM with the same input and output head (6,918 parameters) is the deep-learning baseline.

Sequential objective. Turnover depends on the previous holdings, so weeks cannot be shuffled as independent samples. Each epoch is one full-batch step over the whole chronological training sequence of K weeks, minimizing

$$\mathcal{L} = -\overline{G} + \lambda_r \operatorname{Var}(G) + \lambda_t \overline{T}, \qquad T_k = \|\mathbf{w}_k - \tilde{\mathbf{w}}_k\|_1, \quad (1)$$

where $G_k = \mathbf{w}_k^\top \mathbf{r}_k$ is the gross (pre-cost) return of the target weights $\mathbf{w}_k$ over the week's open-to-open asset returns $\mathbf{r}_k$, $\operatorname{Var}(G)$ the variance penalty over $G_1, \ldots, G_K$, and $\overline{T}$

the mean L1 turnover penalty, with $\tilde{\mathbf{w}}_k = \mathbf{w}_{k-1} \odot (\mathbf{1} + \mathbf{r}_{k-1})/(1 + G_{k-1})$ the drifted previous holdings ($\tilde{\mathbf{w}}_1 = \mathbf{0}$: purchase from cash). λ_r and λ_t weight the risk and turnover penalties; costs enter only the backtester. The four variants are *return* (λ_r=λ_t=0), *risk*, *turnover* and *full*.

Training and selection. AdamW (learning rate 10^{-3}, weight decay 10^{-4}), gradient clipping at 1, at most 120 epochs, early stopping (patience 20) on validation net Sharpe at 10 bps. Penalty strengths are searched with seed 11 over $\lambda_r \in \{1, 5, 20\}$ and $\lambda_t \in \{5\times10^{-4}, 2.5\times10^{-3}\}$. Seeds 29 and 47 reuse the selected strengths and choose only their stopping epoch. Each variant is reported as the average of its three seeds' weights ("ensemble").

Backtester and baselines. All strategies share one backtester. It charges costs on traded notional, solving exactly for the self-financing cost factor $\mu = 1 - c\sum_i |\mu w_i - \tilde{w}_i|$, and charges the initial purchase. The baselines are weekly equal weight; minimum variance and mean–variance (risk aversion 5), both estimated on a 252-day rolling window with 10% covariance shrinkage; the LSTM trained with the full objective; and, as an extra baseline beyond the proposal, a frozen Chronos-2-Synth forecaster [5], pre-trained on synthetic data only, whose 5-day median forecasts are mapped to weights by a fixed softmax (temperature 0.05).

Implementation. The results below come from a native PyTorch [8] implementation; reusing DeepDow's [7] scaler and softmax allocation layer, as proposed, is in progress but not yet used here. The sequential loss and backtester are our own, since DeepDow's losses treat samples independently. Two team members built independent implementations of the training loop as a cross-check. The pipeline passes 53 unit tests (causality of features, split boundaries, cost accounting) and 2,931 automated checks on saved outputs.

4. Preliminary Results (Validation, 2018–2020)

Penalties improve validation Sharpe. Table 1 and Figure 1 show that all three regularized variants beat return-only training at 10 bps (Sharpe: risk 1.10, turnover 0.86, full 1.08; return 0.81). Risk-only and full are close: the ensembles favour risk-only, the seed means favour full (full 1.03 ± 0.07, risk 0.98 ± 0.07, turnover 0.81 ± 0.13, return 0.76 ± 0.12), both within seed variation. The advantage of the full objective is balance: against return-only it cuts volatility from 10.3% to 6.8%, drawdown from 16.8% to 11.8% and turnover from 4.5 to 3.1 (the lowest volatility and turnover among learned models) for 0.8 points of CAGR, and it has the best Sharpe at 25 bps (1.02). A second, independent implementation (1-layer encoder, no dropout, 10 epochs, seeds 42–44, a different λ grid) gives the same seed-mean order (0.998, 0.867, 0.747, 0.717) on the same validation period. The full Transformer also beats every classical baseline on Sharpe, though minimum variance keeps lower volatility (4.6%) and drawdown.

The two penalties interact. Penalizing variance alone doubles turnover (9.0 vs. 4.5), so the risk-only model loses 0.27 Sharpe between 0 and 25 bps, against 0.11 for the full objective. The full objective overtakes risk-only above about 10 bps (Figure 1, right). The turnover-only variant changes little (Sharpe 0.86 vs. 0.81; turnover 4.4 vs. 4.5). Its value appears mainly in combination with the risk term.

Where the gain comes from. Most of the improvement comes from the 2020 sell-off (Figure 2). Between 19 February and 23 March 2020, the full ensemble lost 6.2%, compared with 14.5% for return-only and 14.9% for equal weight. In calmer years it earned less (15.5% vs. 18.7% in 2019). The allocations explain why: the full ensemble holds on average 44% IEF and 19% TLT, compared with 12% and 21% for return-only. This bond tilt relies on Treasuries hedging equities, which held throughout our sample: the daily SPY–IEF correlation was -0.42 in training and -0.46 in validation.

Sensitivity. On the seed-11 search, validation Sharpe rises with λ_r at every λ_t (Table 2). The selected $\lambda_r = 20$ and $\lambda_t = 2.5\times10^{-3}$ are the largest values searched, so the search does not show that they are optimal (see Section 5).

Limitations. Penalties and stopping epochs were selected on the same validation period, so Table 1 overstates out-of-sample performance. That period contains one dominant event (the 2020 crash), and its result rests on a stock–bond hedge that may not hold later. Three seeds measure training noise, not market uncertainty. Two of three return-only seeds selected epoch-1 checkpoints, suggesting the stopping rule partly fits validation noise.

5. Remaining Work

7–15 Nov. Widen the λ_r and λ_t grids beyond their current upper ends. Add walk-forward selection (expanding training windows with several validation folds) so that the chosen penalties do not depend on a single crisis. **16–23 Nov.** Compare variance with downside risk (semivariance or CVaR) and a direct Sharpe loss [1]. Add a regime breakdown by rolling stock–bond correlation. Merge the two implementations. **24–30 Nov.** Analyse the completed test backtest: extend the primary bootstrap (full vs. return, 10 bps) to each penalty vs. return at 0/10/25 bps, with failure cases. Models from the changes above will be secondary analyses; the locked test result stays primary. **1–5 Dec.** Write the final report and package the code.

References

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Method	Net CAGR %	Vol. %	Sharpe	MDD %	Turnover	Sharpe @0 bps	Sharpe @25 bps
Equal weight	9.5	10.4	0.78	−16.8	1.0	0.79	0.77
Minimum variance	5.7	4.6	0.92	−9.0	1.2	0.94	0.88
Mean–variance	3.7	15.8	0.22	−18.0	14.4	0.31	0.08
Chronos-2-Synth (frozen)	9.7	10.4	0.80	−14.9	5.8	0.85	0.72
LSTM, full objective	10.0	9.7	0.88	−14.8	9.7	0.98	0.73
Transformer, return	9.8	10.3	0.81	−16.8	4.5	0.86	0.75
Transformer, risk	10.9	8.4	**1.10**	−13.1	9.0	**1.20**	0.93
Transformer, turnover	10.5	10.5	0.86	−16.2	4.4	0.91	0.80
Transformer, full	9.0	6.8	1.08	−11.8	**3.1**	1.13	**1.02**

Table 1. Validation performance (2018–2020), net of 10 bps per traded dollar unless stated. Sharpe is annualized net excess Sharpe; turnover is annualized L1 turnover (multiples of portfolio value per year). Neural rows are three-seed weight ensembles. Bold marks the best value in a column among learned models. These numbers were also used for model selection and are therefore optimistic.

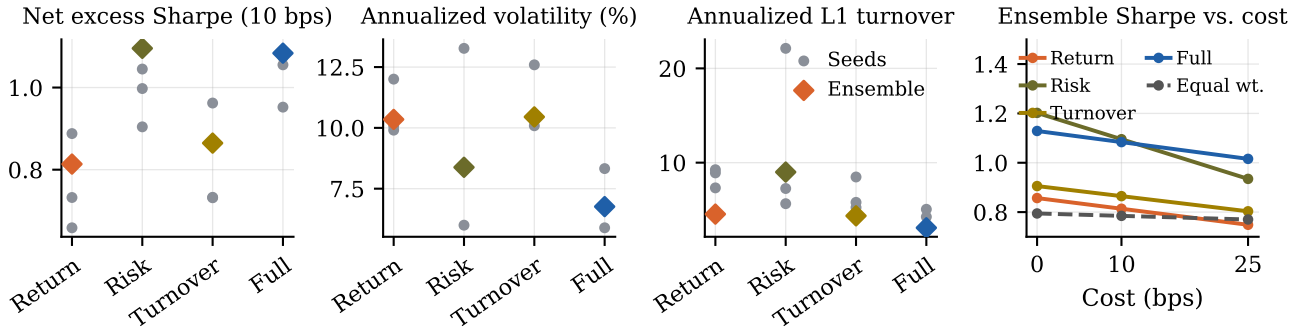


Figure 1. Objective ablations on validation. Left three panels: individual seeds (grey) and three-seed weight ensembles (diamonds) at 10 bps. Right: ensemble Sharpe as costs rise. The risk-only variant leads at zero cost but trades the most and falls behind the full objective above about 10 bps.

	$\lambda_t = 0$	5×10^{-4}	2.5×10^{-3}
$\lambda_r = 0$	0.66	0.68	0.73
$\lambda_r = 1$	0.72	0.74	0.78
$\lambda_r = 5$	0.81	0.83	0.87
$\lambda_r = 20$	0.90	0.89	**0.95**

Table 2. Validation net Sharpe (10 bps) of the Transformer over the penalty grid, seed 11.

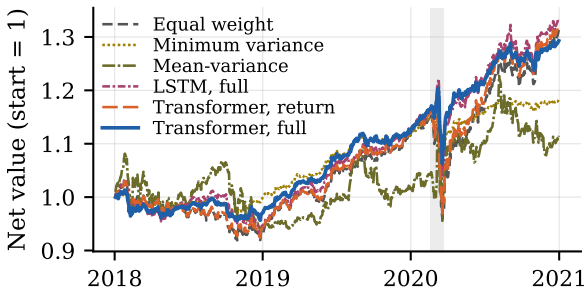


Figure 2. Validation net value at 10 bps. Grey band: 19 Feb–23 Mar 2020 COVID sell-off.

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